

MARICIA ALLEMAN

Senior Data Engineer · Analytics Engineer · BI Developer

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SUMMARY

Hands-on data engineer and analytics builder with 20+ years of experience designing SQL pipelines, ETL/ELT workflows, data warehouse architectures, and Power BI reporting solutions from the ground up. Proven ability to wrangle inconsistent, multi-vendor data sources into governed, reliable models in greenfield environments with little existing infrastructure. Deep technical fluency in SQL Server, Snowflake, Python, Power BI (DAX/Power Query), and Power Automate. U.S. Army veteran with an M.S. in Computer Science and a track record of building what others spec out.

TECHNICAL SKILLS

Databases & Warehousing: SQL Server, Snowflake, UniData, T-SQL, DAX, Power Query, ETL/ELT, Data Modeling, Data Warehousing

BI & Visualization: Power BI (semantic models, row-level security, DAX measures), SSRS, Visual Studio BI

Languages & Automation: Python (Tkinter, file processing, automation), Power Automate, ASP.NET, Git

Platforms & Tools: Ellucian Colleague ERP, OnDataSuite, Eduphoria, Focus SIS, Active Directory, Office 365, SharePoint, Cisco Meraki, Hyper-V

Practices: Data Governance, Data Quality & Validation, FERPA Compliance, ETL Pipeline Design, Greenfield Architecture

EXPERIENCE

Director of District Data Analytics | Ector County ISD — Odessa, TX

2023 – Present

Recruited to build a district-wide data platform from scratch in a greenfield environment — no existing data warehouse, no standardized tooling, and data sourced from a different vendor or system for nearly every reporting area.

- Architected and built ETL pipelines integrating inconsistent, multi-vendor data sources including Focus SIS, OnDataSuite, TEA/TEAL, assessment vendors, and Eduphoria into centralized SQL Server reporting models
- Designed and deployed Power BI semantic models and dashboards supporting attendance, graduation, CCMR, STAAR, TELPAS, and A–F accountability reporting for executive and campus audiences
- Developed CCMR Classification Engine: implemented full TEA-aligned accountability logic, bucket assignments, IBC tier classification, and weighted scoring calculations from raw source data
- Built SDR Sorter — a Python/Tkinter automation tool to process, validate, categorize, and route STAAR, TELPAS, and EOC assessment files; eliminated hours of manual file handling per cycle
- Managed end-to-end State Data Reporting (SDR) and SIRS reconciliation processes including TIDE uploads, student test file extractions, crosswalks, and quality checks
- Developed Python utilities and Power Automate flows for file intake, normalization, naming convention enforcement, and automated routing across systems
- Implemented data governance standards where none existed: data dictionaries, quality checks, lineage documentation, and source-of-truth validation across reporting domains
- Supported Teacher Incentive Allotment (TIA) data compilation, designation logic, validation, and submission documentation in coordination with HR
- Administered Board Goals and District Improvement Plan (DIP) platforms including user provisioning, permissions, and annual rollover configuration

Chief Technology & Information Security Officer | Eastern New Mexico University — Ruidoso 2021 – 2023

- Directed enterprise technology infrastructure: servers, networking, virtualization (Hyper-V), wireless (Cisco Meraki), Active Directory, Office 365, and information security operations
- Built a Power BI application integrating Active Directory and Banner ERP data into a searchable computer inventory lookup — technology used SQL, Power BI, and PDQ

- Developed a Power Automate / Power Apps phone directory solution exporting Mitel Connect data to a mobile-friendly, Teams-embedded Power BI report with multi-format download
- Managed SQL Server, WSUS, Group Policy, and enterprise security operations including disaster recovery planning and business continuity
- Led infrastructure modernization: VLAN configuration, firewall management, wireless expansion, and campus-wide technology replacement planning

System Analyst II | University of Texas Permian Basin

2014 – 2020

- Designed and maintained ETL packages to import 79 extracts from two PeopleSoft databases into a Microsoft SQL Server data warehouse using Visual Studio and SSIS
- Built and maintained SSRS reporting systems and SharePoint-integrated BI solutions supporting institutional compliance and operational decision-making
- Ensured data integrity, quality, and security across enterprise data warehouse systems serving academic and administrative reporting needs
- Developed and customized web-based reporting systems for internal and external compliance requirements

Programmer Analyst / Network Communications Specialist | Odessa College

2009 – 2014

- Administered Ellucian Colleague ERP and WebAdvisor: security provisioning, user access, UniData programs, file dictionaries, and enterprise reporting processes
- Automated ERP user creation and security assignment workflows, significantly reducing manual administration overhead
- Developed and maintained SQL-based institutional reporting solutions and database processes
- Supported enterprise backup systems, disaster recovery, and business continuity operations

PC Support Specialist | City of Odessa

2005 – 2009

- Developed and maintained client/server applications, intranet systems, and SQL-based web applications using ASP.NET across municipal departments
- Maintained SharePoint Server SQL databases and supported business continuity and disaster recovery planning
- Provided hardware, software, and network troubleshooting support across municipal departments

Signal Support Specialist (31U) | U.S. Army

1993 – 2001

- Installed, maintained, and troubleshoot communications systems, signal support equipment, and data distribution systems
- Provided technical training for communications and automation equipment operations; maintained readiness documentation

SELECTED PROJECTS

- **Lone Star Data Platform:** Designed a layered Snowflake architecture using raw, staging, and analytics schemas to centralize district data from disparate sources. Integrated automated Power BI reporting and governed analytics workflows — built without a pre-existing data infrastructure.
- **CCMR Classification Engine:** Engineered full TEA-aligned College, Career, and Military Readiness accountability logic in SQL: bucket assignments, IBC tier classification, and weighted calculations. Powers campus and district Power BI dashboards aligned to A–F accountability requirements.
- **SDR Sorter:** Built a Python/Tkinter desktop application to automate processing, validation, categorization, and routing of STAAR, TELPAS, and EOC assessment files. Replaced a fully manual workflow, reducing processing time and eliminating mis-routing errors.
- **PeopleSoft Data Warehouse:** Designed and implemented ETL packages importing 79 structured extracts from two PeopleSoft databases into a SQL Server data warehouse. Developed SSRS and SharePoint-integrated reporting on top of the warehouse for institutional compliance reporting.
- **Power BI Phone Directory:** Automated export of Mitel Connect Director phone data via Power Automate, stored via Power Apps, and surfaced in a mobile-friendly Power BI report embedded in Microsoft Teams — with multi-format print/download options.

EDUCATION

M.S. Computer Science · University of Texas Permian Basin

GPA 3.8

Thesis: *Augmented Reality Industrial Training Mobile Application Developed with Unity*

B.S. Computer Science · University of Texas Permian Basin

GPA 3.5